Callum Mackenzie

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Experience

Embedded Systems Software Engineer Intern Jan 2027
Tesla AI *Palo Alto, California, USA*

- Incoming on AI Platforms team, supporting Optimus & Tesla AP in January 2027

Software Development Engineer Intern May – Aug 2026
Amazon *Vancouver, BC, Canada*

- Built the **agent harness** for an LLM ticket-triage platform on AWS Bedrock (tool integration via **MCP**, structured output parsing, fallback chains), delivering root-cause analysis for **2,000+ production tickets** across 8 Amazon teams
- Engineered **context and retrieval pipelines** grounding every analysis in team runbooks and **760+ ingested ticket resolutions** (2,800+ RAG retrievals), with source citations on agent responses
- Designed **guardrails for autonomous actions** (confidence gates, default-deny allowlists), enabling the agent to safely post **1,900+ ticket responses** and resolve **90+ tickets end-to-end** without human intervention

Embedded Software Engineer Co-op Jan 2025 – May 2026
General Dynamics *Calgary, AB, Canada / Remote*

- **Saved $20k+ per vehicle platform** by enabling an additional user on existing hardware, eliminating the need for another communications unit
- Developed **3 Linux device drivers and daemons in C/C++** for tactical communications systems spanning **20+ networks of 4-8 distributed nodes**
- Implemented **configurable layer 3/4/5 networking** for communication across distributed embedded systems
- **Reduced CI/CD pipeline runtime by 50%** using build caching and dedicated Docker build environments to eliminate redundant rebuilds
- Built lab-local **AI infrastructure and agents** for testing and development, supporting **20+ engineers** and 3 build pipelines

Education

The University of British Columbia Vancouver, BC, Canada
BSc Computer Science & Statistics (GPA 4.0/4.33) *Sep 2022 – Dec 2027*

Relevant Coursework: Operating Systems, Machine Learning & Data Mining, Computer Vision, Algorithm Design & Analysis, Statistical Learning, Applied Linear Algebra, Computer Graphics, Stochastic Processes

Projects

FITNET - Real-Time Wearable Motion Tracking | *C, Swift, iOS, FreeRTOS, BLE* Feb – Dec 2025

- Built a real-time wearable motion-tracking system using **ESP32-S3 nodes, EMG, and 9-axis IMUs**, with sub-millisecond synchronization across distributed sensors
- Achieved **95% movement-classification accuracy** using DTW-KNN on synchronized EMG and motion data, with on-device orientation estimation using a Mahony AHRS
- Streamed and processed sensor data over optimized **Bluetooth Low Energy** in a SwiftUI/SceneKit iOS app for real-time **3D motion visualization**

Find My Force - RF Signal Classification | *Python, PyTorch, XGBoost, scikit-learn* Mar 2026

- Built an **RF signal classification pipeline** combining CNN-extracted features with statistical signal features for transmitter identification
- Achieved **89% classification accuracy** by ensembling PyTorch CNN, XGBoost, and logistic regression models

Technical Skills

Languages: Python, C++, C, Java, C#, TypeScript, JavaScript, Swift, Rust, Kotlin, R, GLSL
Frameworks & Libraries: PyTorch, scikit-learn, XGBoost, Three.js, Node.js, .NET, SwiftUI, OpenGL, FreeRTOS
Cloud & AI: AWS, AWS CDK, AWS Bedrock, Firebase, MCP, RAG, Supervised & Unsupervised Learning
Systems & Developer Tools: Linux, Git, Docker, Jenkins, CMake, Maven, gdb, Bash